Schottky Barrier Diode

NSVR02HL40MX2W

Schottky Barrier Diodes are Optimized for Very Low Forward Voltage drop and low leakage current and are used in a wide range of dc-dc converter, clamping and protection applications in portable devices. NSVR02HL40MX2W in a X2DFNW2 (0402) miniature package enables designers to meet the challenging task of achieving higher efficiency and meeting reduced space requirements.

Features

- Low Forward Voltage Drop
- Low Reverse Current
- Very High Switching Speed
- 175 °C $T_{J(max)}$-Rated for High Temperature, Mission Critical Applications
- Small Body Outline Dimensions:
  0.039″ x 0.024″ (1.00 mm x 0.60 mm)
- Low Body Height: 0.016″ (0.40 mm)
- Wettable Flank Package for optimal Automated Optical Inspection (AOI)
- NSV Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC-Q101 Qualified and PPAP Capable
- These Devices are Pb-Free, Halogen Free/BFR Free and are RoHS Compliant

Typical Applications

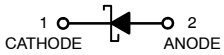
- Buck and Boost dc-dc Converters
- Reverse Voltage and Current Protection
- Clamping & Protection

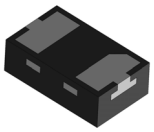
MAXIMUM RATINGS

Rating		Symbol	Value	Unit
Reverse Voltage		V_R	40	V
Forward Current (DC)		I_F	200	mA
Non-Repetitive Peak Forward Surge Current		I_{FSM}	2.0	A
ESD Rating:	Human Body Model Machine Model	ESD	Class 1C Class A	

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

40 V SCHOTTKY BARRIER DIODE





X2DFNW2
CASE 711BG

MARKING DIAGRAM



XX = Specific Device Code
M = Date Code

ORDERING INFORMATION

Device	Package	Shipping†
NSVR02HL40MX2WT5G	X2DFNW2 (Pb−Free)	8000 / Tape & Reel

† For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specification Brochure, BRD8011/D.

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THERMAL CHARACTERISTICS

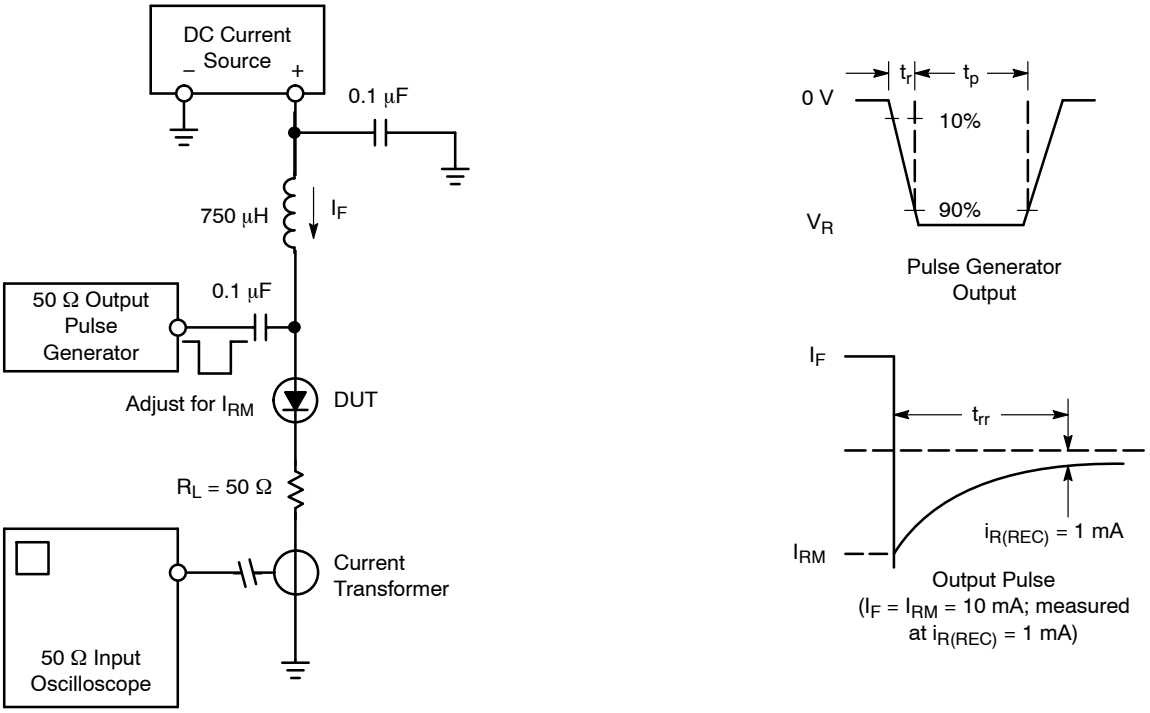
Characteristic	Symbol	Min	Typ	Max	Unit
Thermal Resistance Junction-to-Ambient (Note 1)	$R_{\theta JA}$			400	°C/W
Thermal Resistance Junction-to-Solder Point (Note 1)	$R_{\theta JSP}$			100	°C/W
Junction and Storage Temperature Range	T_J, T_{stg}			−55 to +175	°C

1. Mounted onto a 4 in square FR-4 board 10 mm sq. 1 oz. Cu 0.06” thick single sided. Operating to steady state.

ELECTRICAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)

Characteristic	Symbol	Min	Typ	Max	Unit
Reverse Leakage (V_R = 25 V) (V_R = 40 V) (V_R = 30 V, T_J = 150 °C) (V_R = 40 V, T_J = 150 °C)	I_R		 0.03 0.04 150 200	 0.1 0.5 200 500	μA
Forward Voltage (I_F = 1 mA) (I_F = 10 mA) (I_F = 40 mA) (I_F = 100 mA)	V_F		 0.32 0.43 0.62	 0.38 0.50 0.80 1.20	V
Total Capacitance (V_R = 1.0 V, f = 1 MHz)	CT		 2.0	 5.0	pF
Reverse Recovery Time (I_F = I_R = 10 mA, I_R = 1.0 mA)	t_{RR}		 1.5	 4.0	ns

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.



1. DC Current Source is adjusted for a Forward Current (I_F) of 10 mA.
2. Pulse Generator Output is adjusted for a Peak Reverse Recovery Current I_{RM} of 10 mA.
3. Pulse Generator transition time << t_{rr}.
4. IR(REC) is measured at 1 mA. Typically 0.1 x I_{RM} or 0.25 x I_{RM}.
5. t_p » t_{rr}

Figure 1. Recovery Time Equivalent Test Circuit

Share Feedback
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TYPICAL CHARACTERISTICS

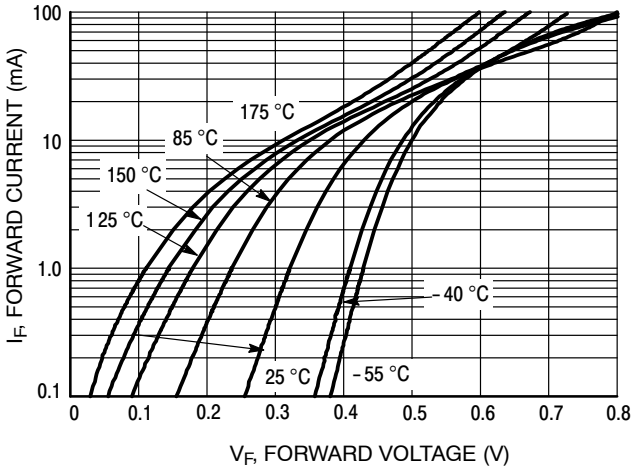


Figure 2. Typical Forward Voltage

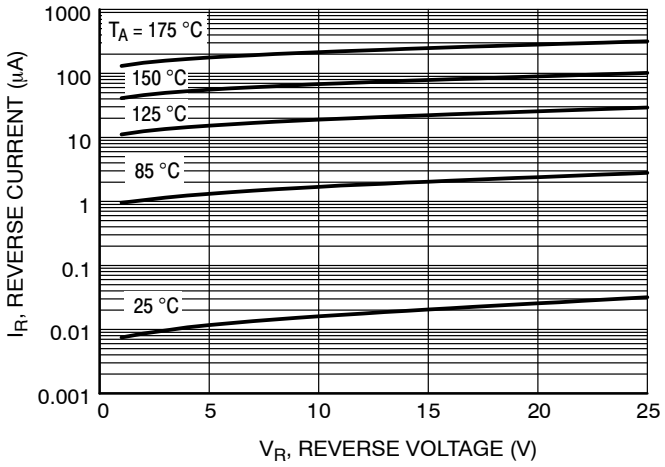


Figure 3. Reverse Current versus Reverse Voltage

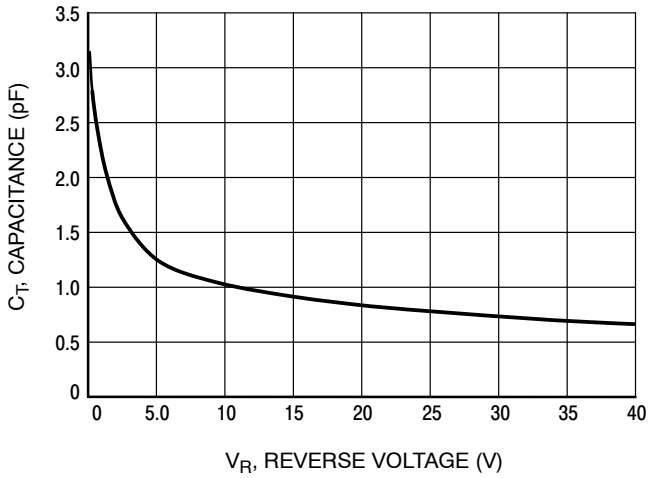


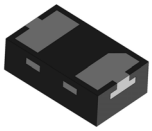
Figure 4. Typical Capacitance

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REVISION HISTORY

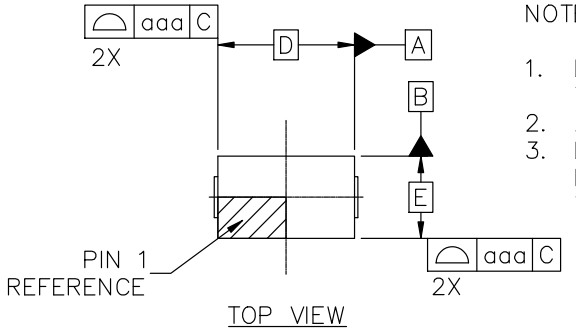
Revision	Description of Changes	Date
0	Production Version Released.	1/15/2021
1	Update of the front page case outline 3D model.	11/20/2025
2	Typo – Figure 2 IF from μA to mA UOM.	12/10/2025

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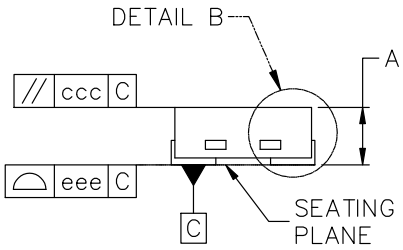


X2DFNW-2 1.00x0.60x0.37, 0.65P
CASE 711BG
ISSUE F

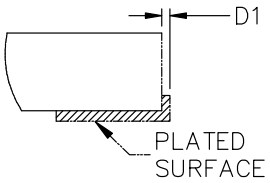
DATE 06 NOV 2025



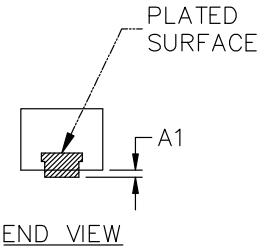
TOP VIEW



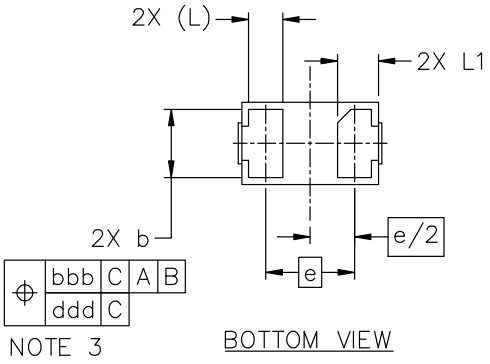
SIDE VIEW



DETAIL "B"
SCALE 2:1



END VIEW



BOTTOM VIEW

NOTES:

1. DIMENSIONING AND TOLERANCING CONFORM TO ASME Y14.5−2018.
2. ALL DIMENSION ARE IN MILLIMETERS.
3. DIMENSION b APPLIES TO THE PLATED TERMINAL AND IS MEASURED BETWEEN 0.15 AND 0.30 FROM THE TERMINAL TIP.

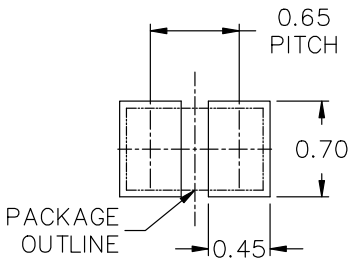
MILLIMETERS			
DIM	MIN	NOM	MAX
A	0.34	0.37	0.40
A1	~	~	0.05
b	0.45	0.50	0.55
D		1.00 BSC	
D1	~	~	0.05
E		0.60 BSC	
e		0.65 BSC	
L		0.22 REF	
L1	0.24	0.28	0.34
TOLERANCE FORM & POSITION			
aaa		0.05	
bbb		0.10	
ccc		0.05	
ddd		0.05	
eee		0.05	

GENERIC MARKING DIAGRAM*



XX = Specific Device Code
M = Date Code

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, "G", may or not be present. Some products may not follow the Generic Marking.



RECOMMENDED MOUNTING FOOTPRINT*

* FOR ADDITIONAL INFORMATION ON OUR Pb−FREE STRATEGY AND SOLDERING DETAILS, PLEASE DOWNLOAD THE onsemi SOLDERING AND MOUNTING TECHNIQUES REFERENCE MANUAL, SOLDERRM/D.

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DESCRIPTION:	**X2DFNW-2 1.00x0.60x0.37, 0.65P**		**PAGE 1 OF 1**

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